



System Agent

Live System Monitoring Project

The goal was to build something similar to how network management systems work in telecom, where devices report their status to a central server and raise alarms when something goes wrong. Instead of network devices, the agents run on computers and report CPU, RAM, and disk usage in real time.

The project started as a simple agent that collected metrics and printed them in a terminal. It grew into a full monitoring system with a central server, a SQLite database that persists all readings, and a web dashboard accessible from the browser. The dashboard has 4 tabs: a live view per machine, a multi machine comparison view, an alarm management tab, and a full history tab with statistics and CSV export.

The alarm system works by automatically raising an incident when CPU, RAM, or Disk crosses a threshold, assigning it a severity level of Minor, Major, or Critical. Each incident stays open until the metric recovers or an engineer manually resolves it, which is similar to how alarm management is handled in real telecom networks.

The system was tested across two machines at the same time, a Windows machine and a Linux VM running AlmaLinux inside VirtualBox, both sending data to the same server simultaneously.

Built by Nafisa Menacere, **GitHub:** https://github.com/Nafisacere/system_agent

Log in Page

The dashboard is protected by a login screen

**System Agent**
Monitoring Dashboard

Sign in

Enter your credentials to continue

USERNAME

Username

PASSWORD

Password

Sign in

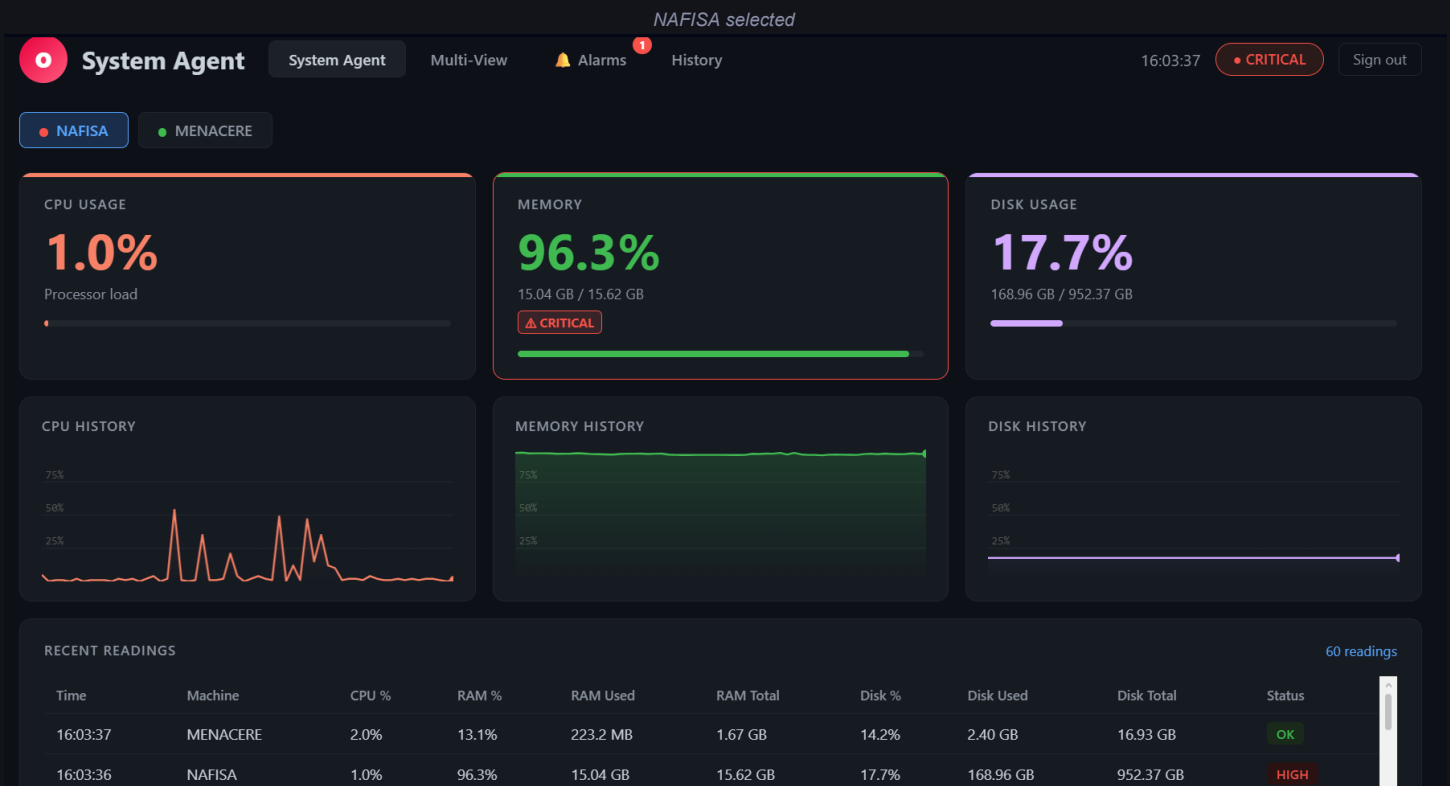
Navigation Bar

Stays visible across all pages with four tabs. The Alarms tab shows a live badge with the number of open incidents



System Agent Tab

Live KPI cards, real time history charts and recent readings table. NAFISA (Windows) and MENACERE (Linux VM) both reporting at the same time. The tabs on top show which machine is selected and its current status.



(NAFISA's RAM hit 96.3% the Memory card turns red and the header switches from LIVE to CRITICAL automatically)



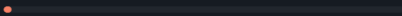
NAFISA

MENACERE

CPU USAGE

2.0%

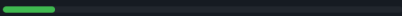
Processor load



MEMORY

13.1%

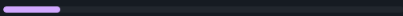
223.2 MB / 1.67 GB



DISK USAGE

14.2%

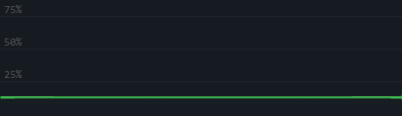
2.40 GB / 16.93 GB



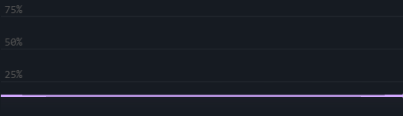
CPU HISTORY



MEMORY HISTORY



DISK HISTORY



RECENT READINGS

60 readings

Time	Machine	CPU %	RAM %	RAM Used	RAM Total	Disk %	Disk Used	Disk Total	Status
16:03:00	MENACERE	2.0%	13.1%	223.2 MB	1.67 GB	14.2%	2.40 GB	16.93 GB	OK
16:03:00	NAFISA	4.0%	95.5%	14.93 GB	15.62 GB	17.7%	168.95 GB	952.37 GB	HIGH

Multi-View Tab

Compares all machines side by side across CPU, Memory, and Disk. Overview chart combines all metrics in one view

NAFISA MENACERE

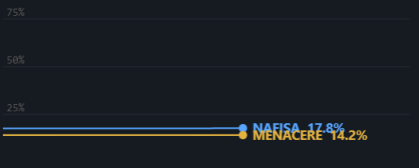
CPU USAGE



MEMORY USAGE

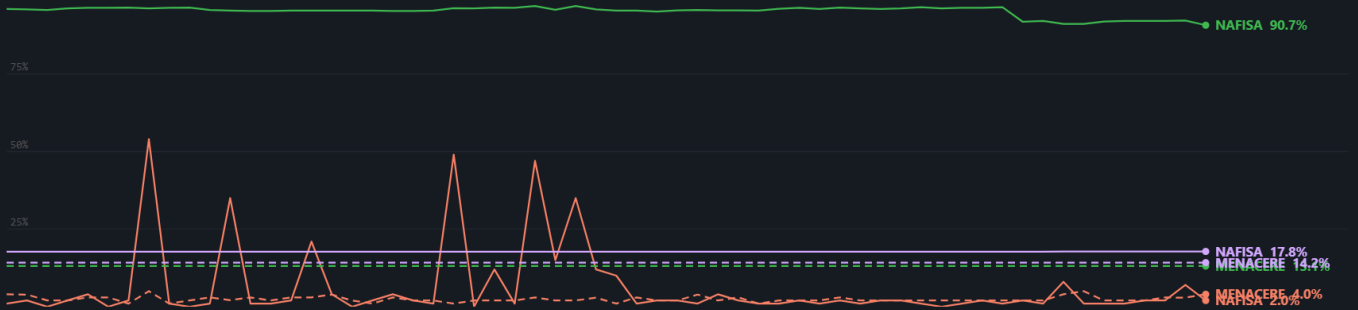


DISK USAGE



OVERVIEW

CPU Memory Disk | NAFISA MENACERE



Alarms Tab

Raises an incident when a metric gets too high (Minor $\geq 60\%$, Major $\geq 75\%$, Critical $\geq 90\%$). You review it and click Resolve when it's fixed.

System Agent

System Agent

Multi-View

Alarms ¹

History

16:05:13

CRITICAL

Sign out

CRITICAL (OPEN)

1

MAJOR (OPEN)

0

MINOR (OPEN)

0

RESOLVED (TOTAL)

16

INCIDENT LOG 17 incidents

AllOpenResolvedAll Machines

ID	Severity	Machine	Metric	Value	Threshold	Raised At	Duration	Status	Action
#17	CRITICAL	NAFISA	RAM	91.3%	$\geq 90\%$	6/10/2026 16:05:08	3s	OPEN	Resolve
#1	MAJOR	NAFISA	RAM	85.3%	$\geq 75\%$	6/9/2026 13:44:42	59s	RESOLVED	6/9/2026 13:45:42
#2	MAJOR	NAFISA	RAM	86%	$\geq 75\%$	6/9/2026 13:45:43	7s	RESOLVED	6/9/2026 13:45:50
#4	MINOR	NAFISA	CPU	60%	$\geq 60\%$	6/9/2026 13:48:17	2s	RESOLVED	6/9/2026 13:48:19
#5	MAJOR	NAFISA	CPU	76%	$\geq 75\%$	6/10/2026 10:52:40	2s	RESOLVED	6/10/2026 10:52:43
#3	MAJOR	NAFISA	RAM	85.7%	$\geq 75\%$	6/9/2026 13:45:52	21h 13m	RESOLVED	6/10/2026 10:59:33
#7	MINOR	NAFISA	CPU	66%	$\geq 60\%$	6/10/2026 11:00:27	3s	RESOLVED	6/10/2026 11:00:31
#8	MINOR	NAFISA	CPU	74%	$\geq 60\%$	6/10/2026 11:03:58	3s	RESOLVED	6/10/2026 11:04:01
#9	MINOR	NAFISA	CPU	60%	$\geq 60\%$	6/10/2026 11:07:42	6s	RESOLVED	6/10/2026 11:07:49

History Tab (Uptime & Stats)

Shows uptime and avg/min/max stats per machine, filterable by machine and time range, with history table and CSV export

System Agent

System Agent

Multi-View

Alarms ¹

History

16:23:06

CRITICAL

Sign out

MENACERE

146h 7m

15,123 readings recorded

Active — last seen 16:23:04

NAFISA

147h 16m

5,101 readings recorded

Active — last seen 16:23:03

STATISTICS

All MachinesAll time

MENACERE

CPU Avg	CPU Min	CPU Max
2.3%	0%	38.3%
RAM Avg	RAM Min	RAM Max
13.5%	10.7%	14.4%
Disk Avg	Disk Min	Disk Max
14.2%	14.2%	14.2%

Based on 15,124 readings

NAFISA

CPU Avg	CPU Min	CPU Max
7.9%	0%	99%
RAM Avg	RAM Min	RAM Max
79.7%	63.5%	97.1%
Disk Avg	Disk Min	Disk Max
17.1%	16.5%	18%

Based on 5,101 readings

HISTORY 20,129 readings — All Machines · All time

All MachinesAll time

Download CSV

History table (View when you scroll down)

HISTORY19,604 readings — All Machines · All time

All Machines

All time

Download CSV

Time	Machine	CPU %	RAM %	RAM Used	RAM Total	Disk %	Disk Used	Disk Total
6/10/2026 16:06:18	MENACERE	2%	13.1%	223.2 MB	1.67 GB	14.2%	2.40 GB	16.93 GB
6/10/2026 16:06:17	MENACERE	2%	13.1%	223.2 MB	1.67 GB	14.2%	2.40 GB	16.93 GB
6/10/2026 16:06:17	MENACERE	1%	13.1%	223.2 MB	1.67 GB	14.2%	2.40 GB	16.93 GB
6/10/2026 16:06:16	MENACERE	3.9%	13.1%	223.2 MB	1.67 GB	14.2%	2.40 GB	16.93 GB
6/10/2026 16:06:16	NAFISA	2%	91.3%	14.27 GB	15.62 GB	17.8%	169.38 GB	952.37 GB
6/10/2026 16:06:16	MENACERE	2%	13.1%	223.2 MB	1.67 GB	14.2%	2.40 GB	16.93 GB
6/10/2026 16:06:15	MENACERE	1%	13.1%	223.2 MB	1.67 GB	14.2%	2.40 GB	16.93 GB
6/10/2026 16:06:15	MENACERE	2%	13.1%	223.2 MB	1.67 GB	14.2%	2.40 GB	16.93 GB
6/10/2026 16:06:14	MENACERE	2%	13.1%	223.2 MB	1.67 GB	14.2%	2.40 GB	16.93 GB
6/10/2026 16:06:14	MENACERE	4.6%	13.1%	223.2 MB	1.67 GB	14.2%	2.40 GB	16.93 GB
6/10/2026 16:06:13	MENACERE	2%	13.1%	223.2 MB	1.67 GB	14.2%	2.40 GB	16.93 GB
6/10/2026 16:06:13	NAFISA	10%	91.7%	14.33 GB	15.62 GB	17.8%	169.38 GB	952.37 GB

1

2

3

Next →

NAFISA: Windows Agent

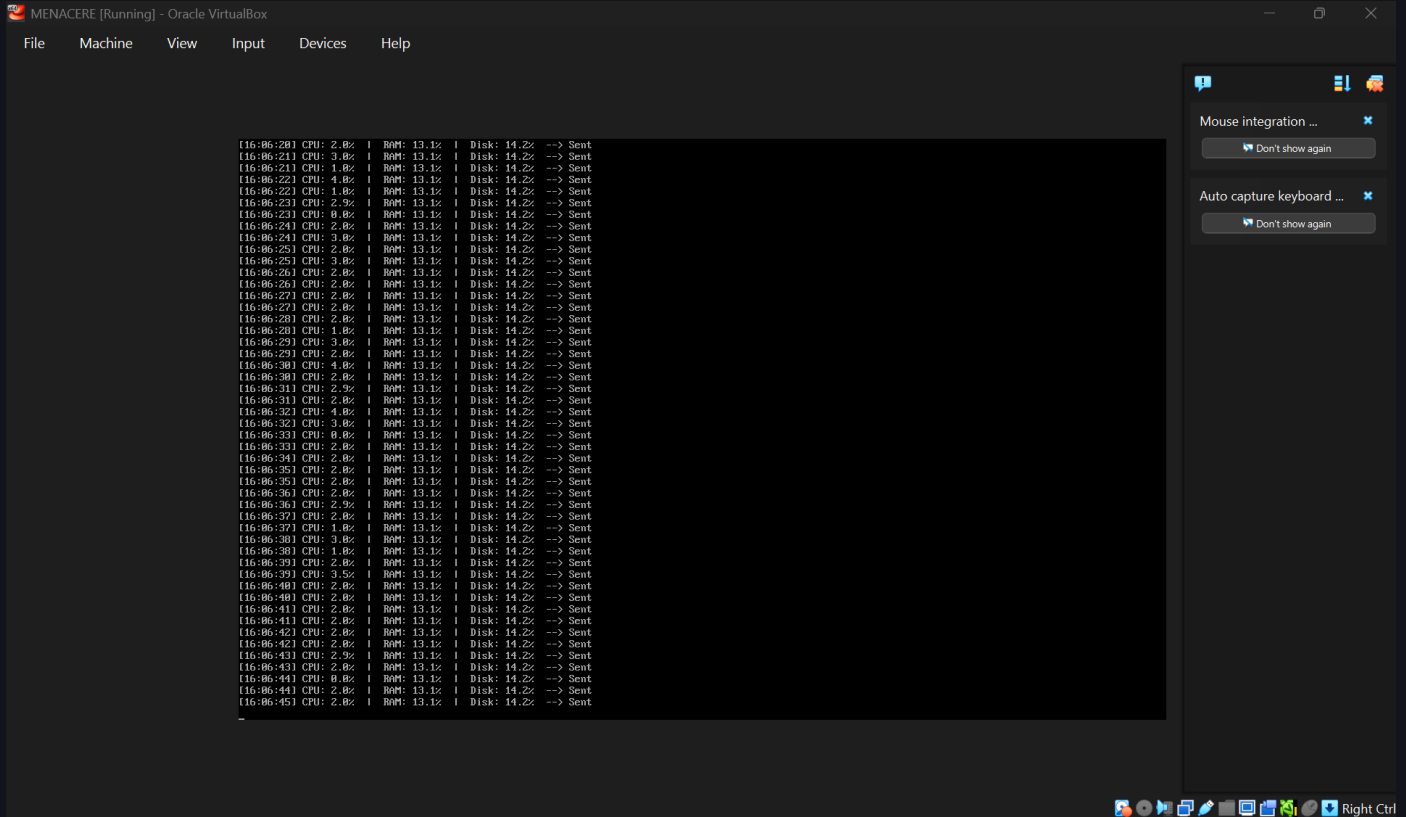
NAFISA is a Windows machine running the agent. It collects metrics and sends them to the server over the network.

```
SYSTEM AGENT
Machine      : NAFISA
Sending to: localhost:9001 (UDP)

[15:33:53] CPU: 13.0% | RAM: 68.0% | Disk: 17.8% --> Sent
[15:33:56] CPU: 49.0% | RAM: 68.3% | Disk: 17.8% --> Sent
[15:33:59] CPU: 18.0% | RAM: 68.0% | Disk: 17.8% --> Sent
[15:34:02] CPU: 58.0% | RAM: 68.7% | Disk: 17.8% --> Sent
[15:34:05] CPU: 76.0% | RAM: 69.6% | Disk: 17.8% --> Sent
[15:34:09] CPU: 46.0% | RAM: 70.6% | Disk: 17.8% --> Sent
[15:34:11] CPU: 43.0% | RAM: 70.6% | Disk: 17.8% --> Sent
[15:34:14] CPU: 29.0% | RAM: 70.3% | Disk: 17.8% --> Sent
[15:34:17] CPU: 44.0% | RAM: 70.2% | Disk: 17.8% --> Sent
[15:34:20] CPU: 12.0% | RAM: 69.4% | Disk: 17.8% --> Sent
[15:34:23] CPU: 15.0% | RAM: 69.4% | Disk: 17.8% --> Sent
[15:34:26] CPU: 12.0% | RAM: 69.4% | Disk: 17.8% --> Sent
[15:34:28] CPU: 16.0% | RAM: 69.3% | Disk: 17.8% --> Sent
[15:34:31] CPU: 21.0% | RAM: 68.4% | Disk: 17.8% --> Sent
[15:34:34] CPU: 18.0% | RAM: 68.3% | Disk: 17.8% --> Sent
[15:34:37] CPU: 21.0% | RAM: 68.4% | Disk: 17.8% --> Sent
[15:34:40] CPU: 13.0% | RAM: 68.0% | Disk: 17.8% --> Sent
[15:34:43] CPU: 33.0% | RAM: 68.1% | Disk: 17.8% --> Sent
[15:34:46] CPU: 27.0% | RAM: 68.0% | Disk: 17.8% --> Sent
```

MENACERE: Linux VM (VirtualBox)

A separate Linux machine used to prove the system can monitor more than one computer at a time. It collects metrics and sends them to the server over the network.



Server Startup & Live Log

Server listens on UDP port 9001 and serves the dashboard on port 9000. Once agents are connected, readings from NAFISA and MENACERE come in simultaneously

```
[Server] Database ready: metrics.db
MONITORING SERVER
Agents send data to : port 9001 (UDP)
Dashboard          : http://localhost:9000/
Database           : metrics.db
Alarm thresholds   : MINOR>=60% MAJOR>=75% CRITICAL>=90%

[Server] Listening on UDP port 9001
[16:22:36] NAFISA      | CPU:    2.0% | RAM:  93.4% (14.59 GB / 15.62 GB) | Disk:  17.8% | CRITICAL [from 127.0.0.1]
[16:22:39] NAFISA      | CPU:    0.0% | RAM:  93.4% (14.59 GB / 15.62 GB) | Disk:  17.8% | CRITICAL [from 127.0.0.1]
[16:22:42] NAFISA      | CPU:    0.0% | RAM:  93.4% (14.59 GB / 15.62 GB) | Disk:  17.8% | CRITICAL [from 127.0.0.1]
[16:22:45] NAFISA      | CPU:    1.0% | RAM:  93.3% (14.58 GB / 15.62 GB) | Disk:  17.8% | CRITICAL [from 127.0.0.1]
[16:22:46] MENACERE    | CPU:    0.0% | RAM:  11.7% (200.2 MB / 1.67 GB) | Disk:  14.2% | OK       [from 127.0.0.1]
[16:22:47] MENACERE    | CPU:   12.3% | RAM:  11.8% (201.6 MB / 1.67 GB) | Disk:  14.2% | OK       [from 127.0.0.1]
[16:22:47] MENACERE    | CPU:    4.8% | RAM:  11.8% (201.6 MB / 1.67 GB) | Disk:  14.2% | OK       [from 127.0.0.1]
[16:22:48] NAFISA      | CPU:    1.0% | RAM:  93.5% (14.61 GB / 15.62 GB) | Disk:  17.8% | CRITICAL [from 127.0.0.1]
[16:22:48] MENACERE    | CPU:    1.0% | RAM:  11.8% (201.6 MB / 1.67 GB) | Disk:  14.2% | OK       [from 127.0.0.1]
[16:22:48] MENACERE    | CPU:    2.9% | RAM:  11.8% (201.6 MB / 1.67 GB) | Disk:  14.2% | OK       [from 127.0.0.1]
[16:22:49] MENACERE    | CPU:    5.0% | RAM:  11.8% (201.6 MB / 1.67 GB) | Disk:  14.2% | OK       [from 127.0.0.1]
[16:22:49] MENACERE    | CPU:    2.0% | RAM:  11.8% (201.6 MB / 1.67 GB) | Disk:  14.2% | OK       [from 127.0.0.1]
[16:22:50] MENACERE    | CPU:    2.9% | RAM:  11.8% (201.6 MB / 1.67 GB) | Disk:  14.2% | OK       [from 127.0.0.1]
[16:22:50] MENACERE    | CPU:    2.0% | RAM:  11.8% (201.6 MB / 1.67 GB) | Disk:  14.2% | OK       [from 127.0.0.1]
[16:22:51] NAFISA      | CPU:    2.0% | RAM:  94.9% (14.83 GB / 15.62 GB) | Disk:  17.8% | CRITICAL [from 127.0.0.1]
```